CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

M.Tech. (EC) (C.S.E.) SEM – I Examination. 2010-11

EC - 703 Digital System Design

Date: 05.01.11, Wednesday   Time: 10:30 a.m to 01.30 p.m   Maximum Marks: 70

Instructions:

1. **Attempt all questions.**
2. **Answer to the two sections must be written in separate answer books.**
3. **Figures to the right indicate *full* marks.**
4. **Make suitable assumptions and draw neat figures wherever required.**

SECTION-I

Q-1 (a) List the various advance features of CPLD compared to SPLD. 3

(b) Explain the concept of concurrency and delta delay in VHDL. 4

(c) A Mealy sequential network with four output variables is realized using 22V10. What is the maximum number of input variables it can have? Can any Mealy network with these numbers of inputs and outputs be realized with a 22V10? Justify. 2

(d) A sequential network has one input (x) and one output (z). The network examines groups of four consecutive inputs and produces output z=1 if the input sequence 0101 *or* 1001 occurs. The network resets after every four inputs. Find the Mealy state graph. 2

Q-2 (a) Explain control state graph of 4 bit parallel binary divider. 5

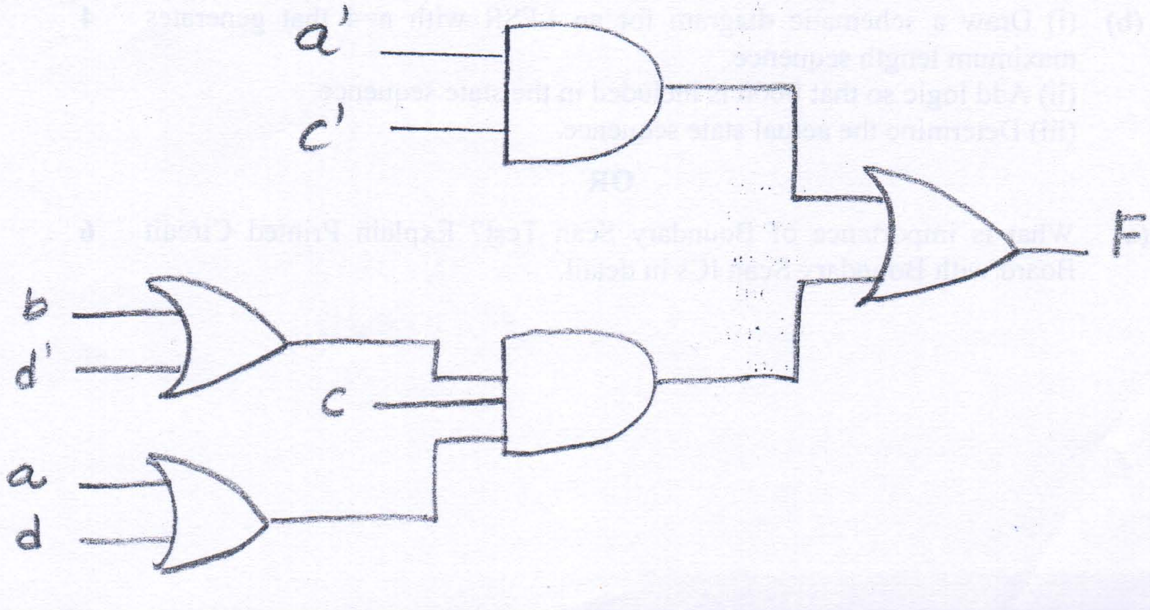
(b) Write a model for modulo 10 counter using VHDL. 4

(c) Explain the different components of SM Chart. 3

OR

Q-2(a) Draw a block diagram of a 4-bit signed serial parallel multiplier. Explain operation of it and also describe control state graph. 7

(b) What do you mean by hazards? Find static 1 and static 0 hazards in the following network. For each hazard, specify the values of the input variables and which variable is changing when hazard occurs. Also specify the order in which the gate outputs must change. 5



Q-3 (a) Suppose, you work on a specification for a system with some digital parameters. Each parameter has Min, Typ and Max columns. In what column would you put a Setup time and a Hold time? Justify your answer. **3**

(b) Explain the SM chart for Dice game and implement it using a PLA and flip flops. **9**

OR

Q-3(a) Draw mealy state machine for detecting a sequence "1001". Convert a state graph to an SM chart. **6**

(b) Using the method of map entered variables, use 4 variable maps to find a minimum sum of product expression. Also verify it with 5 variable kmap. **6**

$F(A,B,C,D,E)=\sum m(0,4,5,7,9)+\sum d(6,11)+E(m_1,m_{15})$

$F(A,B,C,D,E)=\sum m(0,4,6,13,14)+\sum d(2,9)+E(m_1,m_{12})$

SECTION-II

Q-4(a) Which are the basic reasons to move from PROM to PLA to PAL? **4**

(b) Define the following terms: **2**
Bridging faults, Error

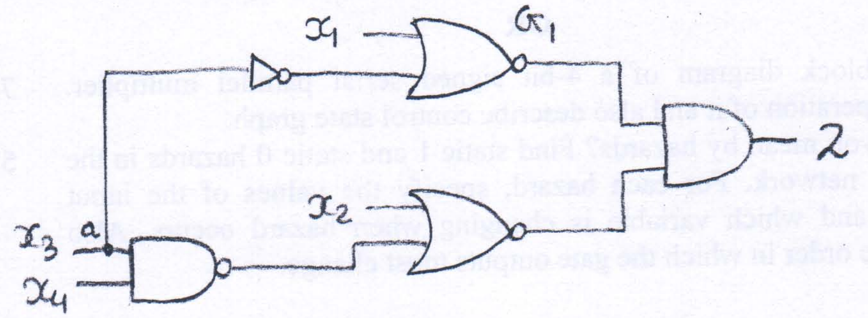
(c) Describe a finite state machine that will detect three consecutive coin tossed (of one coin) that results in heads. **3**

(d) What is the difference between testing and verification? **2**

Q-5 (a) For the given combinational circuit, **8**

(i) Determine the output function for the
-AND bridge between inputs of gate G1
-The multiple fault {x_3 s-a-1, x_2 s-a-0}

(ii) Find the set of all tests that detect the fault x_1 s-a-0.



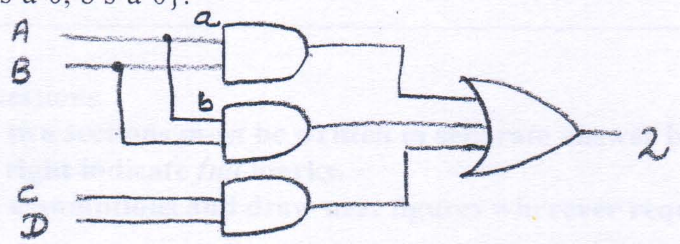
(b) (i) Draw a schematic diagram for an LFSR with n=4 that generates maximum length sequence. **4**
(ii) Add logic so that 0000 is included in the state sequence.
(iii) Determine the actual state sequence.

OR

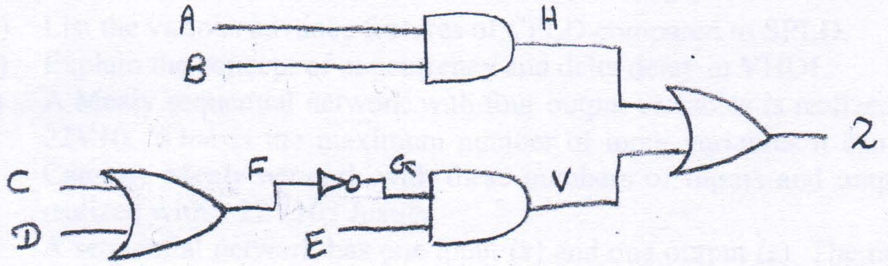
Q-5 (a) What is importance of Boundary Scan Test? Explain Printed Circuit Board with Boundary Scan ICs in detail. **6**

(b) For the given combinational circuit, 6

1. Find the set of all tests that detect the fault a s-a-0.
2. Find the set of all tests that detect the fault b s-a-0.
3. Find the set of all tests that detect the multiple fault {a s-a-0, b s-a-0}.



Q-6 (a) Apply fault collapsing theory to the circuit shown below. Write down the reduced test set for the same. 6



(b) Describe the complete design flow for programmable logic with all necessary figures and details. 6

OR

Q-6 (a) Compare the different programming technologies used in PLD. 6

(b) Find a minimum-row PLA table to implement the following sets of functions. 6

$F_1(A,B,C,D)=\sum m(4,5,10,11,12)$.

$F_2(A,B,C,D)=\sum m(0,1,3,4,8,11)$.

$F_3(A,B,C,D)=\sum m(0,4,10,12,14)$.